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Microfluidic Protein Preconcentrator Using a Microchannel-Integrated Nafion Strip: Experiment and Modeling

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We propose a simple microfluidic device for protein preconcentration based on the electrokinetic trapping principle. It comprises a narrow Nafion strip that is simply cut from a commercial membrane and is integrated into a molded poly(dimethylsiloxane) (PDMS) microfluidic structure using a guiding channel. Mechanically clamping the PDMS/Nafion assembly with a glass substrate results in a rapid prototypable, leak-tight, and easily disposable device. Our device preconcentrates negatively charged fluorescent proteins located at the anodic microfluidic compartment side of the Nafion strip within a few minutes and up to a concentration factor of 10^4. Moreover, we present a numerical study of the preconcentration effect by solving the coupled Poisson, Nernst−Planck, and Navier−Stokes equations for our type of device, which provides microscopic insight into the mechanism of preconcentration. The electrical field across the ion-permselective Nafion generates concentration polarization, i.e., ion depletion at the anodic side and ion enrichment at the cathodic side for both types of ions, with a local excess of mobile positive ions in the depleted concentration polarization zone, inducing a nonequilibrium electrical double layer in close proximity to the Nafion membrane. A voltage difference applied over the anodic compartment is used to generate the electrophoretic flow velocity of the negatively charged tracer biomolecules. This, in combination with the electroosmotic flow in the opposite direction, which originates from the fixed charges on the channel walls and the induced space charge near the membrane, provides the basis for the local preconcentration of the negative tracer biomolecules.

Microfluidic devices have many potential advantages over larger sized biomedical analysis systems; for example, they enable integration of multiple biochemical processing steps onto a single monolithic device.[1,2] Applications, such as biomolecule purification, accumulation, separation, reaction, and amplification, have already been demonstrated on a miniaturized scale.[3−6] Microscale separation devices often can only analyze a small fraction of the available sample, which can pose a limit to the overall detection sensitivity, especially for the detection of low-concentration species. In proteomics, this problem is aggravated by the fact that targeted tracer molecules, for example, cytokines and biomarkers, are present only in very low concentrations (nanomolar to picomolar range). Furthermore, no amplification technique exists for proteins and peptides, such as the polymerase chain reaction (PCR) for nucleic acids. Thus, a high-efficiency protein preconcentrator able to purify low-concentration biomarkers from microliter sample volumes is extremely useful for implementation of high-sensitivity diagnosis in the microfluidic chip format.[7]

There are many techniques available for sample preconcentration in microfluidic devices. These include field-amplified sample stacking,[8] isotachophoresis,[9] isoelectric focusing,[10,11] chromatographic trapping,[12,13] electrokinetic trapping,[14−26] tem-

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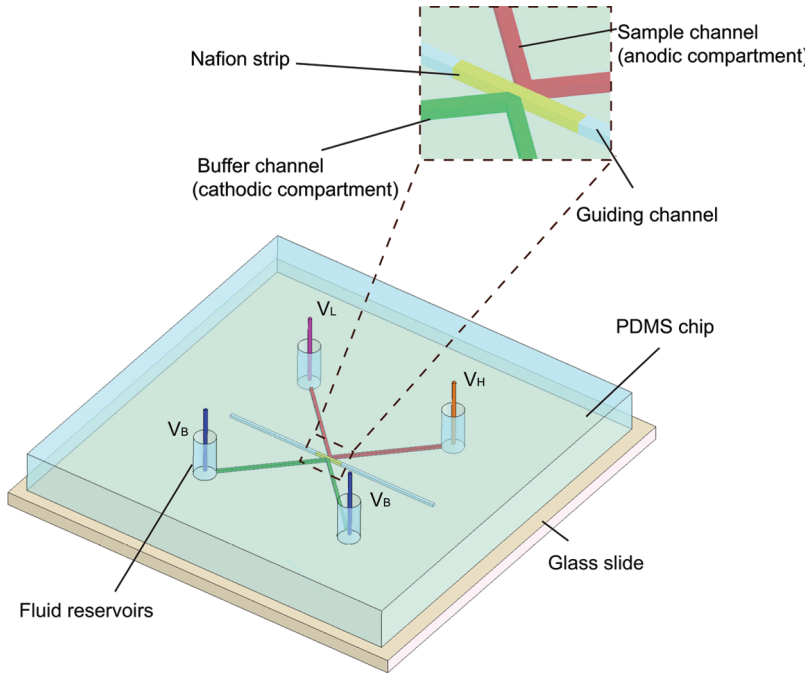

Figure 1. Schematic illustration of the protein preconcentration chip in PDMS with an integrated Nafion strip, with the definition of the voltages applied to the reservoirs during operation. The zoom shows in more detail the guiding channel for embedding the Nafion strip that separates the sample from the buffer channels.

perature-gradient focusing,[27] etc. Electrokinetic trapping is particularly interesting, as it can be used for any charged biomolecule and allows the use of simple fluid-handling protocols. However, it generally calls for specially fabricated nanoporous membranes[14−16,20,23,28,29] or nanofluidic channels.[6,17,18,21,22,30] Using advanced microfabrication techniques, the group of J. Han[18,22] proposed a high-efficiency nanofluidic channel-based preconcentrator demonstrating a million-fold concentration of charged proteins. Kim et al.[19] successfully concentrated fluorescein isothiocyanate (FITC)−bovine serum albumin (BSA) by utilizing spontaneously formed nanochannels at a bonded poly(dimethylsiloxane) (PDMS)−glass interface. Though cumbersome fabrication steps were avoided in this device, an extreme control of the PDMS−glass bonding operation was required. Alternatively, nanoporous polymer membranes were in situ formed inside microchannels, thereby avoiding complex clean room nanofabrication processes.[14,16,28,29] In situ polymerization on-chip allows easy variation of the chemical composition and surface charge of the nanoporous material, but may require the application of dedicated washing procedures and can result in a lack of control of the nanoporosity, when compared to optimized commercial ion-exchange membrane fabrication processes. Recent work using commercially available ion-exchange membranes (Nafion 115) for the electrokinetic capture of micrometer-sized beads in a microfluidic channel was reported by Lindberg et al.[31] Nafion is a highly ion-permselective membrane normally used in proton exchange membrane (PEM) fuel cells.[32−34] Recently, we demonstrated a monolithic microfluidic PEM fuel cell by integrating a narrow Nafion strip into a molded PDMS structure.[35] Our design overcame many of the complex fabrication and assembly problems related to the membrane integration in a microfluidic structure and formed the technological basis for the present study.

Here we propose a simple microfluidic device for protein preconcentration based on the electrokinetic trapping principle. Our preconcentrator comprises a narrow Nafion strip that is simply cut from a commercial membrane and is integrated into a molded PDMS microfluidic structure using a specially designed guiding channel. Our device offers biomolecule preconcentration within a few minutes up to a concentration factor of 10^4. For microscopic understanding of the phenomenon, we present a numerical simulation by simultaneously solving the Nernst−Planck, Poisson, and Navier−Stokes equations for our type of device.

DESIGN AND WORKING PRINCIPLE

The preconcentrator is based on a glass substrate that is mechanically clamped with a PDMS chip which has two microfluidic channels and a particular guiding channel for inserting the Nafion strip. The latter, having fixed negative charges and a pore mean size on the order of the electrical double layer (EDL) thickness, behaves as a cation-permselective membrane.[34] Figure 1 shows the Nafion-strip-separated microfluidic sample and buffer

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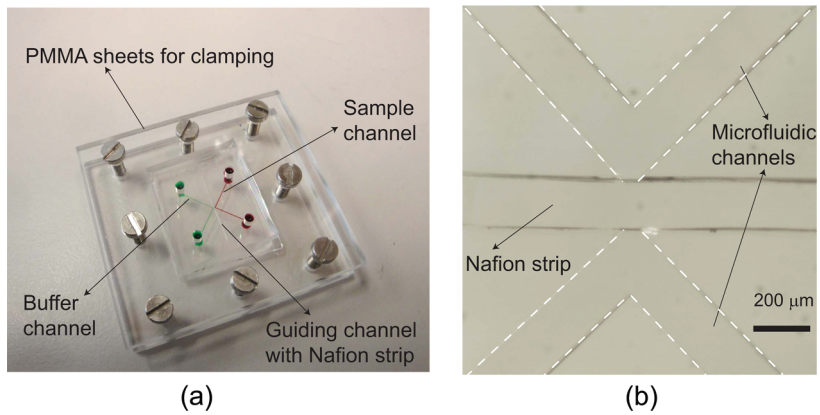

Figure 2. (a) Photograph of the PDMS preconcentration chip with its mechanical clamping setup. (b) Optical micrograph of the microfluidic circuit with an integrated Nafion strip.

channels, forming the anodic and cathodic sides of the device, respectively. The fluidic reservoirs of the cathodic compartment (buffer channel) are connected to $V_B = 0$ V, while the remaining two reservoirs of the anodic compartment (sample channel) have positive voltages V_H and V_L, respectively. The electrical field across the ion-permselective Nafion generates concentration polarization (CP), i.e., ion depletion at the anodic side and ion- enrichment at the cathodic side, with a local excess of mobile positive ions in the depleted CP zone, very close to the membrane, inducing a nonequilibrium EDL. The voltage difference ($V_H - V_L$) imposed over the anodic compartment is used to generate the electrophoretic flow velocity of the negative tracer biomolecules. This, in combination with the electroosmotic flow (EOF) induced by the fixed charges on the anodic channel walls (EOF of the first kind) and the EOF induced by the space charge of the nonequilibrium EDL (EOF of the second kind), provides the basis for the enrichment of the tracer biomolecules (see further text).

EXPERIMENTAL SECTION

Materials. The PDMS prepolymer set (Sylgard 184) was purchased from Dow Corning Inc. (Midland, MI). The float glass slide was purchased from Menzel GmbH (Braunschweig, Germany). BSA conjugated with Alexa Fluor 647 (AF−BSA) was obtained from Invitrogen (Basel, Switzerland), and this was used as the tracer protein in the following experiment. The Nafion strip was cut from a larger Nafion 117 membrane (thickness 175 μm, DuPont). The mean pore size is 1−3 nm according to different literature reports.[36,37] The fixed volume charge concentration of the Nafion membrane is around −1200 mol/m^3.[37] Phosphate-buffered saline (PBS) (1×, 10 mM, pH 7.4) solution was prepared by diluting a 10× PBS concentrated solution purchased from Sigma-Aldrich Chemie GmbH (Buchs, Switzerland). The PBS−Tween (PBST) solution (pH 7.4) was prepared by mixing 0.5% (v/v) Tween-20 (Sigma-Aldrich) with 1× PBS. A PBS−BSA 1% solution was obtained by diluting 1% (w/v) BSA (Sigma-Aldrich) in 1× PBS. Deionized (DI) water of resistivity 18.2 MΩ cm was obtained from the Ecole Polytechnique Fédérale de Lausanne's (EPFL's) Center of Micro- and Nanotechnology. Pt wires of diameter 0.25 mm were obtained from Goodfellow Cambridge Ltd. (Huntingdon, U.K.).

Microfluidic Device Fabrication. The master structure for replication of the PDMS chip was made from a ~170 μm thick negative SU-8 photoresist on a silicon substrate using standard processing, as reported before.[35] By utilizing two razor blades, we were able to cut a Nafion strip of ~200 μm width. The Na-form Nafion was prepared as follows:[38] After standard pretreatment steps,[35] the strip was stirred in 1 M NaOH at room temperature for 24 h. Then it was rinsed several times with DI water until pH 7. The strip was finally kept in 0.1 M NaCl solution for at least one day. The Na-form Nafion is less swollen and has a higher mechanical strength than the usual H-form Nafion.[39] After the PDMS structure was demolded, it was flipped, and the integration of the Nafion strip was realized thanks to the guiding channel. Using a diamond tip, a glass slide with dimensions of 28 mm × 28 mm was cut and mechanically assembled with the PDMS microstructure using poly(methyl methacrylate) (PMMA) support sheets (3 mm thickness), enabling leakage-free functioning. The preconcentrator chip was finalized by placing four Pt wires as electrodes into the four fluidic reservoirs. The PDMS−glass preconcentrator chip with its mechanical clamping system is shown Figure 2a. The overall dimensions of the PMMA support are 45 mm × 50 mm. Figure 2b shows in more detail a top view of the Nafion strip flanked by the two symmetric microfluidic channels.

Protocols and Measurement Setup. For preconditioning, any impurities in both the sample and buffer channels were first removed by introducing PBST solution into the channels using syringes. Then 1× PBS buffer solution was loaded into the buffer channel, while the sample channel was filled with PBS−BSA 1% solution for blocking the nonspecific binding of the AF−BSA molecules on the channel walls. After 5 min, the latter solution was replaced by 1× PBS solution. Finally, the AF−BSA in 1× PBS solution was loaded into the sample channel with various concentrations (60 pM to 60 nM).

To activate the electrokinetic migration of the protein, first, the Pt electrodes in the two reservoirs of the buffer channel were

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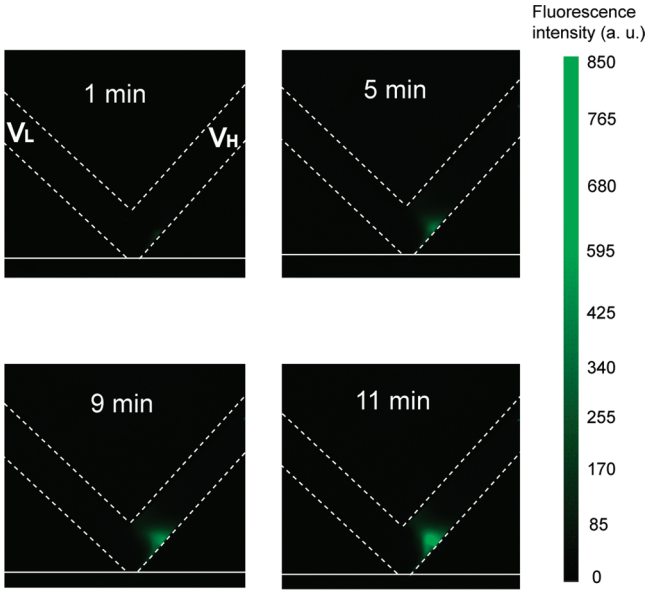

Figure 3. Fluorescence micrographs captured during preconcentration of a 6 nM AF−BSA sample after application of a voltage difference of 10 V ($V_H = 15$ V, $V_L = 5$ V) for 1, 5, 9, and 11 min. Both buffer channel reservoirs (not shown) were kept at 0 V.

connected to zero potential; at the same time, a bias voltage, V_L or V_H, was applied across the sample channel through the other two Pt electrodes. Fluorescence micrographs of the microchannel region juxtaposing the Nafion strip were captured using an Axiovert S100 (Zeiss) inverted microscope equipped with a CCD camera, ORCA-C4742-95ER (Hamamatsu, Shizuoka Prefecture, Japan). The microscope is equipped with a mercury short arc lamp, HBO (Osram), and the appropriate filter set for Alexa Fluor 647 (red). The exposure time used for capture of the fluorescence signal was 2 s, and the software AquaCosmos (Hamamatsu) was used for fluorescence image analysis.

EXPERIMENTAL RESULTS

The most significant experimental parameters involved in the preconcentration are the applied voltage across the membrane and the voltage difference ($V_H - V_L$) in the anodic microfluidic compartment, as well as the tracer protein concentrations. Figure 3 shows the fluorescence images obtained at time frames of 1, 5, 9, and 11 min for a 6 nM sample when exposed to a voltage difference of 10 V over the anodic compartment ($V_H = 15$ V, $V_L = 5$ V). A fluorescence spot becomes visible in the lower right part of the sample channel, the intensity of which increases with increasing time. This preconcentrated plug does not span the complete width of the channel.

To quantify the preconcentration factor, we measured the fluorescence intensity of the preconcentrated plug as a function of time for three different AF−BSA initial concentrations (60 pM, 6 nM, and 60 nM). Each experimental data point and error bar shown in Figure 4 corresponds to the average fluorescence intensity and variance, respectively, corresponding to a sampling area of 16 pixels around the maximum. In the case of the 6 nM AF−BSA concentration, the fluorescence intensity reaches after 10 min the intensity level of a 6 μM AF−BSA reference solution filling the channel, implying a preconcentration factor of $\sim 10^3$. Similarly, the chip is able to effectively preconcentrate a low

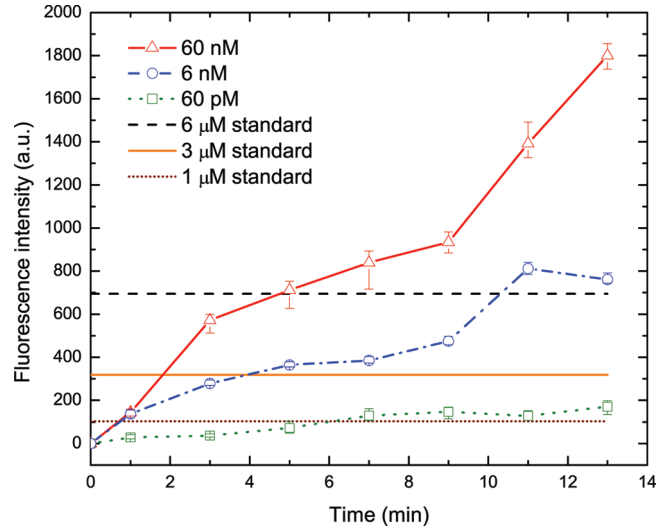

Figure 4. Time dependence of the maximum fluorescence intensity of the preconcentrated plug for AF−BSA concentrations of 60 nM, 6 nM, and 60 pM with the applied electrical potential difference of 10 V ($V_H = 15$ V, $V_L = 5$ V). Also the fluorescence intensity from standard concentration samples (1, 3, and 6 μM) is shown.

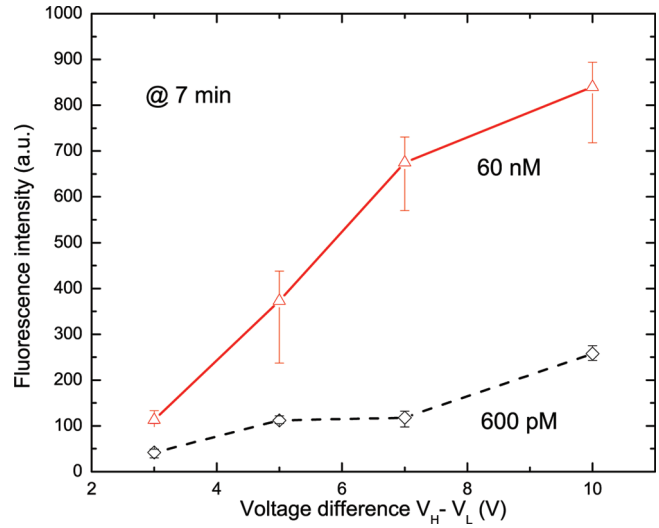

Figure 5. Maximum fluorescence intensity of the preconcentrated AF−BSA plug at a control time of 7 min for initial concentrations of 60 nM and 600 pM as a function of the voltage difference applied over the anodic compartment (keeping $V_L = 5$ V).

AF−BSA concentration of 60 pM by a factor $\sim 2 \times 10^4$ within 7 min, indicating the high potential of our preconcentrator device.

We also measured the fluorescence intensity of the AF−BSA preconcentrated plug as a function of the voltage difference $V_H - V_L$ for a specific time of 7 min (see Figure 5). The fluorescence intensity of the preconcentrated plug increases with the voltage difference for both the 60 nM and 600 pM AF−BSA concentrations. One should note that our preconcentration voltages are of the lowest values reported yet due to the chosen geometrical parameters of the Nafion strip, holding promise for portable applications. In principle, the preconcentration effect could be considerably enhanced when the potential difference $V_H - V_L$ is further increased. However, we found a Nafion membrane deformation and rapid failure (after a few minutes) when applying voltages V_H, $V_L > 20$ V. The reason for this phenomenon could be a low tolerance of the Nafion membrane

to Joule heating.[31] Also, changes in the Nafion membrane during the experiment could be a factor affecting the time-dependent slope of the fluorescence intensity profiles of Figure 4.

THEORETICAL BACKGROUND

In general, the electrokinetic motion of charged molecules in a microfluidic channel is at the basis of transport, separation, and enrichment of ionic species via the complex interplay of (i) electrophoresis of the positive and negative ions of the background electrolyte, (ii) electrophoresis of the tracer biomolecules, and (iii) electro-osmosis of the background electrolyte driven by the charge imbalance in the EDL close to a solid−liquid interface.[14,21,28,40] In this section, we start by introducing the essential features of the EDL, EOF, and ion permselectivity of the porous membrane and then explain their influence on the numerical analysis.

When a nanoporous ion-permselective membrane is placed between two fluidic compartments and an external electrical field applied over it, CP develops, which is a complex of effects related to the formation of concentration gradients of ionic species in the electrolyte solution adjacent to the ion-permselective membrane interface.[40,41] As the field is switched on, the co-ions (here, anions) move away from the anodic interface toward the anode, but their local withdrawal cannot be compensated by the few co-ions which actually remain inside the membrane due to its cation selectivity (co-ion exclusion). Thus, the co-ion concentration decreases locally in the solution adjacent to the anodic interface, and so does the counterion concentration to maintain local electroneutrality. At the opposite, that is, a cathodic interface where counterions leave the membrane, the electrolyte concentration correspondingly increases. This cathodic diffusion boundary layer results in an enriched CP zone. However, the polarized ion current through the membrane creates a positive charge imbalance very close to the membrane interface, where diffusive transport is limited.

In our work, we combined the transient Nernst−Planck equations for both types of buffer ions with a proper description of the momentum transfer of the fluid, as expressed by the Navier−Stokes equation, and linked it with the electrical potential by applying the Poisson equation.[42−44] The Nernst−Planck equation describes the conservation of mass for ionic species in a fluid medium in the presence of an applied electrical field. In principle, the transport of positive ($i = 1$) and negative ($i = 2$) buffer ions and the tracer molecules ($i = 3$) can be described by this equation:

$$\frac{\partial c_i(t)}{\partial t} = \nabla\left(D_i \nabla c_i(t) + z_i c_i(t) D_i \frac{F}{RT} \nabla \phi(t)\right) - \nabla c_i(t) \cdot \mathbf{u}(t) \tag{1}$$

where c_i, D_i, and z_i are the molar concentration of ionic species i and its corresponding diffusion coefficient and corresponding valence, respectively. F is the Faraday constant, and R and T represent the molar gas constant and temperature, respectively. ϕ is the electrical potential, and $\mathbf{u}$ is the fluid velocity. This equation describes the flux of ions under the influence of both an ionic concentration gradient $\nabla c_i(t)$ and an electrical field $E = -\nabla\phi$.

Poisson's equation is used to describe the relation of the electrical potential and the local concentration distribution of buffer ions and charged tracer molecules:

$$\nabla^2 \phi = -\frac{\rho_{\text{fix}}}{\varepsilon_0 \varepsilon_r} - \frac{F}{\varepsilon_0 \varepsilon_r} \sum_i z_i c_i(t) \tag{2}$$

where ρ_{fix} is the volumetric fixed charge density of the membrane and ε_0 and ε_r are the vacuum permittivity and the relative dielectric constant, respectively.

Moreover, the Navier−Stokes equation and the continuity equation for an incompressible fluid are incorporated into the model, which describes the motion of the fluid:

$$\rho \frac{\partial \mathbf{u}(t)}{\partial t} + \rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \eta \nabla^2 \mathbf{u} - F \sum_i z_i c_i(t) \nabla \phi \tag{3}$$

$$\nabla \cdot \mathbf{u} = 0$$

where$\nabla p = 0$ indicates the absence of external hydrostatic pressure and ρ and η are the density and viscosity of the water-based fluid. The Navier−Stokes equation should be employed with the electrical body force term to incorporate the interaction between the local electrical field and ion concentration gradients, as indicated in the third term on the right in eq 3.

The coupled equations above suggest that solving the mathematical model of a preconcentration system is highly complex and that a numerical method is essential to provide a physical explanation of the phenomena that are at the basis of the preconcentration effect. Numerical modeling of preconcentration systems has been done before, leading to a theoretical understanding of the interplay of CP and electro-osmotic flow effects. However, the full numerical solution of eqs 1−3 requires an extremely fine mesh (a few nanometers) to correctly describe EDL-based effects, which is the reason why mostly micrometer-scale fluidic systems were simulated, to stay within computational resource limitations. We have used the software COMSOL Multiphysics (version 3.5a) to solve the differential equations set for a two-dimensional microfluidic system of centimeter size, taking a fine mesh close to the membrane to describe CP-related charge imbalance effects and taking a slip velocity boundary condition at the microfluidic channel walls to mimic the EOF of the first kind and thereby gain computing power, as shown in the next section.

NUMERICAL SIMULATION

Because of the model complexity, several specific assumptions and conditions were taken into account.

(i) The Brinkman equation for the porous medium (for example, the Nafion membrane) in the absence of external hydrostatic pressure was adopted for describing the flow inside

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the membrane, taking into account the electrical body force from the mobile ions:[45]

$$\frac{\rho}{\varepsilon_p}\frac{\partial \mathbf{u}(t)}{\partial t} + \frac{\eta}{\kappa}\mathbf{u} = \frac{\eta}{\varepsilon_p}\nabla^2\mathbf{u} - nF\sum_i z_i c_i(t)\nabla\phi \quad (4)$$

$$\nabla \cdot \mathbf{u} = 0$$

where $0 < \varepsilon_p < 1$ is the porosity and κ the permeability of the membrane. The correction factor n originates from the reduced velocity due to the electric double layer overlap effect within the pores; we used a value of $n = 0.1$ in our simulations.[14,46,47]

(ii) Because of the existence of the EDL due to the fixed charge on the interface between the microchannel and electrolyte solution, EOF of the first kind is generated by applying an electrical field over the microchannel. However, in the small region close to the membrane−microchannel interface, because of the CP phenomena, a secondary EDL due to the induced space charge develops. To be more precise, near the membrane, the effective EDL, which generates locally an EOF of the second kind, results from the secondary EDL, complemented by the primary (quasi-equilibrium) EDL. To take into account the two kinds of EOF simultaneously and to reduce the complexity of the calculation process, EOF of the first kind was modeled as originating from an effective velocity slip of the liquid at the microchannel walls on the basis of the thin EDL assumption. The slip velocity is given by the Helmholtz−Smoluchowski formula[40,48,49]

$$u_{\text{first EOF}} = -\frac{\varepsilon\zeta}{\eta}E_t \quad (5)$$

where E_t is the tangential electrical field just outside the double layer, ε and η are the permittivity and viscosity of the liquid (both assumed constant), and ζ is the ζ potential. This approach permits avoidance of the use of a very fine mesh to resolve EDL-based effects in the anodic microchannel. Since the EOF of the first kind is implemented into the model in the form of the boundary slip, the fixed surface charge on the microchannel walls is no longer considered.[50]

(iii) Another assumption is that we did not consider separate concentrations of the negative tracer biomolecules and the anions, as, due to the concentrations used, the former have little impact on the CP phenomena and flow velocity, even in the preconcentrated state.

Equations 1−4 were solved using the following boundary conditions: For the Nernst−Planck equation, the ion flux normal to the microchannel walls is zero and the ion concentrations at the four edges are kept constant. For the Poisson equation, fixed electrical potentials are assumed at the four edges of the microchannel; 5 and 15 V are applied on the left and right edges of the anodic compartment, respectively, while the two edges of the cathodic compartment are connected to the ground. Slip boundary conditions are assumed on the channel walls for the fluid velocities in the Navier−Stokes equation, these values being obtained from eq 5. External pressure is assumed to be zero at all edges. The other parameters used in the calculation are listed in Table 1. We performed the simulation by assuming that the current is mainly carried by the ions of the electrolyte, the concentration of the tracer biomolecules being negligible with respect to the electrolyte ionic concentrations. An electrolyte ionic concentration of 1 mM was used (for both positive and negative ions), corresponding to a Debye length of 10 nm.[47,51] Moreover, an extremely fine mesh of 12 nm was employed at the microchannel−porous membrane interface to resolve the induced space charge effect. During the operation of the device, as a depletion zone locally forms in the anodic compartment close to the membrane, the Debye length dramatically increases there, in favor of even better numerical solution accuracy. Solving the coupled equations proved to be a highly nontrivial task, and a convergent solution was reached if $\rho_{fix}/F < 5-10$ mM. We stress that the aim of our model is not to present a quantitative one-to-one correspondence between experiment and theory, but instead to provide us with basic insight into the complex physical mechanisms that are at the basis of the preconcentration effect.

Table 1. Parameters Used in the COMSOL Simulation

diffusion coefficient of a positive/negative ion in the electrolyte[a]	1.5×10^{-9} m^2/s
diffusion coefficient of a positive/negative ion in the membrane[a]	2×10^{-10} m^2/s
membrane porosity, ε_p[a]	0.28
membrane permeability, κ[a]	10^{-18} m^2
assumed ion concentration (positive and negative) in the electrolyte solution, C_0	1 mM
assumed fixed volumetric charge concentration of the membrane	1 mM

[a] The values were taken and estimated from refs 37, 52, and 53 and represent the physical properties of a porous membrane.

SIMULATION RESULTS AND DISCUSSION

Figure 6a shows the simulated concentration of the positive and negative ions of the electrolyte both within the ion-selective membrane and at the anodic and cathodic compartment side. The calculation was done along the solid line of the inset shown in the lower-left corner of the figure, after application of the external voltage (V_L = 5 V, V_H = 15 V) for 1, 7, and 15 min. The concentration of cations in the membrane is higher than that of the anions for the three cases, which is a direct consequence of the permselectivity of the membrane. Moreover, the ionic concentrations in the anodic compartment strongly decrease in time for both types of ions, while this is the opposite in the cathodic compartment (CP effect). As shown in Figure 6b, very close to the membrane interface there is a local surplus of mobile positive ions. This result essentially points to the presence of an induced space charge region over a distance within the channel

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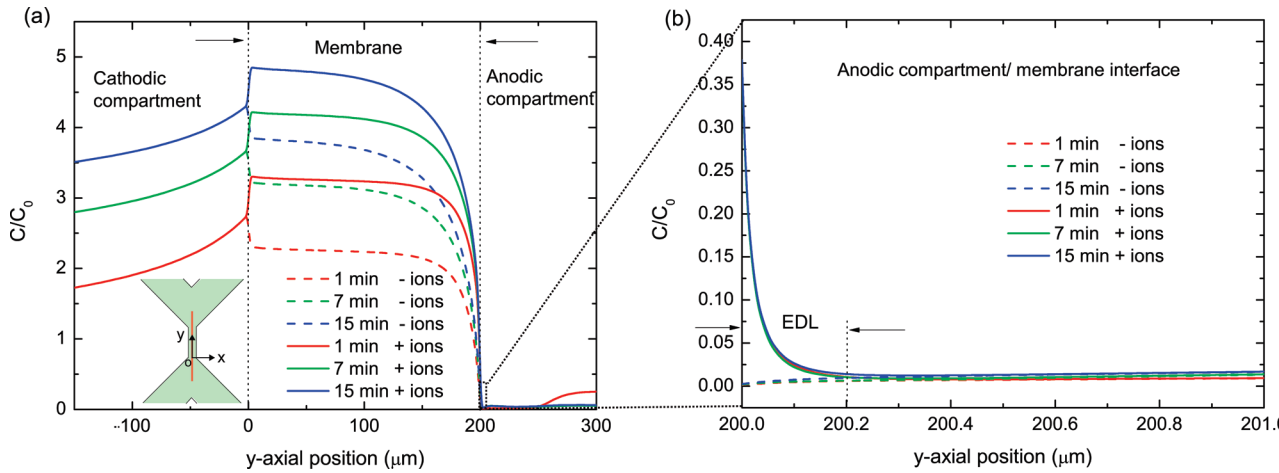

Figure 6. (a) Distributions for monovalent positive (full lines) and negative (dashed curves) ions of the background electrolyte after application of the external voltage ($V_L = 5$ V, $V_H = 15$ V) for 1, 7, and 15 min. The calculation was done along the solid line shown in the inset at the lower left. (b) Zoom on the ion distributions of the background electrolyte in a region close to the membrane, showing charge imbalance and formation of the EDL.

of about 200 nm. Such a secondary EDL, as already suggested in ref 40, is the origin of the EOF of the second kind in the anodic compartment, when a potential difference is applied over the latter. The dynamic change in the local ion concentration, hence in the resistivity, also results in a redistribution of the local electrical potential and field. This type of behavior is in agreement with the experimental measurement of the local potential distribution in the anodic compartment of a nanochannel-based preconcentration device.[17]

Figure 7a shows the simulated streamlines of the fluid in the anodic compartment and membrane area after application of the external voltage ($V_L = 5$ V, $V_H = 15$ V) for 15 min. The normalized arrows indicate the flow direction. The tangential component of the electrical field originating from the potential difference ($V_L - V_H$) across the anodic compartment provides the local driving force on the space charge of the EDL near the membrane, inducing the EOF of the second kind. As the latter is only created on the membrane side of the microchannel, a vortexlike flow pattern is induced. The arrows shown in Figure 7b represent the electrical field strength in an area defined by the dashed rectangle in Figure 7a. Due to the ion depletion effect close to the membrane, the highest field strength is obtained in the region of the main vortex, creating locally an electrical field maximum, inducing an enhanced electrophoretic velocity on the negatively charged tracer biomolecules.[54] This leads to a driving force toward the anodic compartment side with potential V_H. Thanks to the delicate balance between the electrophoretic and electro-osmotic driving forces, the negatively charged tracer molecules will be stacked just right of the field maximum in the anodic compartment, as experimentally observed in Figure 3. Moreover, the topology of the experimental zone of concentration indicates the presence of vortexlike structures indeed. Figure 7c is a plot of the simulated flow profile in the anodic microchannel along the solid line in Figure 7a after application of the external voltage ($V_L = 5$ V, $V_H = 15$ V) for 1, 7, and 15 min. After 1 min, a predominantly flat velocity profile is observed across the channel, mainly caused by the EOF of the first kind, taken as 3 μm/s in the model. On the basis of our experimental parameters, the BSA-coated microchannel under the applied electrical field can provide an EOF in the range of 5−10 μm/s. However, to reach convergence in the simulation, we applied this smaller EOF in the microchannel via the Helmholtz−Smoluchowski slip condition. For increasing times, the ionic depletion zone is created and the secondary EDL and the vortex form, inducing locally a translational motion of the fluid (EOF of the second kind), as evidenced by the parabolic flow profiles. Our simulation results indicate that both types of EOFs are responsible for generation of the fluid velocity in the preconcentration device, while the EOF induced by the nanopores of the membrane is found to be 2 orders of magnitude smaller than the fluid velocity in the microchannel, and hence does not significantly contribute.

It is also interesting to compare our preconcentration device with published preconcentration devices (see Table 2), where our device scores particularly well when considering affordability and portability of lab-on-a-chip applications. First, our novel device is attractive thanks to the straightforward use of the commercial membrane, which avoids complex or in situ micro- and nanofabrication. Second, our limit of detection is as low as tens of picomolar, competitive with other work. Third, the fast preconcentration (10^3−10^4-fold within several minutes) using only a low operation voltage (<15 V) holds potential for future portable applications. It is worthwhile to notice that, in our device, as well as those of others, preconcentration occurred on the anodic side of the ion-selective membrane−nanochannel interface and benefits from the coeffect of the first and space-charge-induced EOF of the second kind. Preconcentration was also reported on the cathodic side of an ion-permselective membrane, but here the primary EOF distorted the phenomenon, and optimized preconcentration was found in this case for a noncharged membrane.[14]

8. CONCLUSION AND OUTLOOK

We have developed a simple protein preconcentrator using PDMS rapid prototyping technology and an original ion-permselective membrane assembly technique. Using low voltages (<15 V), we were able to achieve a preconcentration factor of 10^3−10^4

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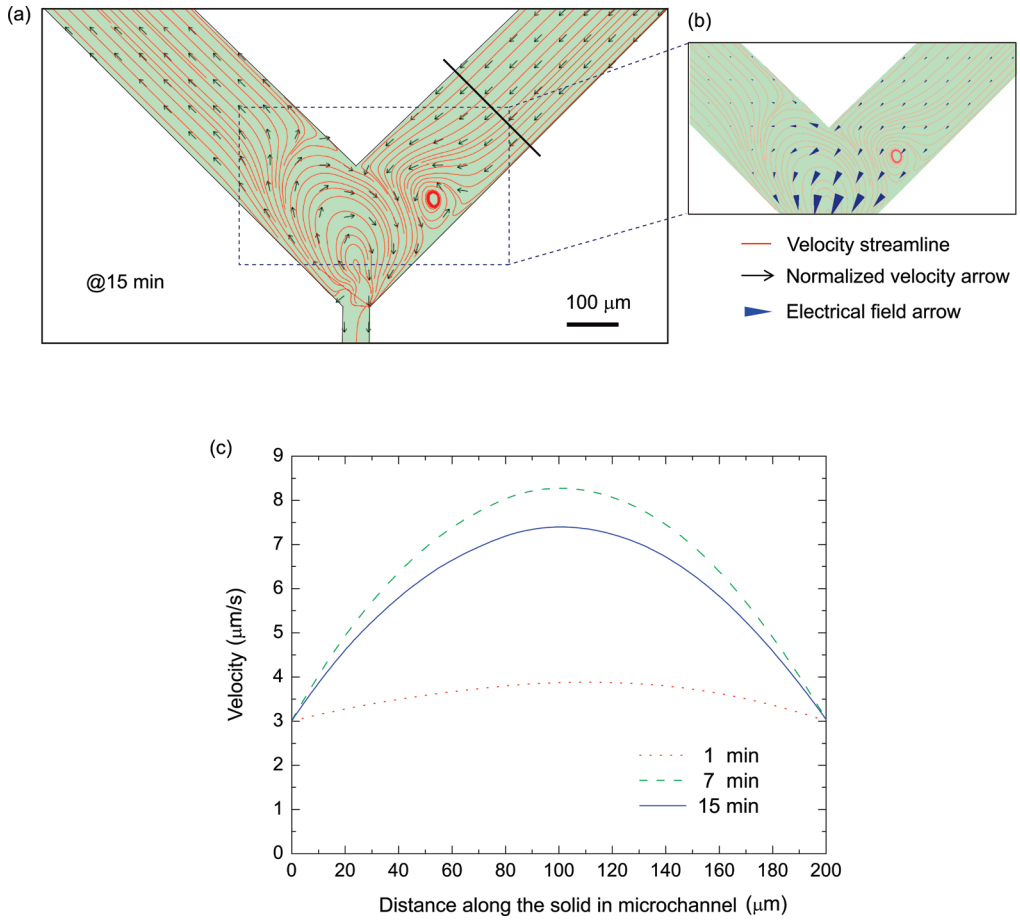

Figure 7. (a) Simulated fluid velocity streamlines (uniform density plot) with normalized arrows indicating the flow direction after application of the external voltage ($V_L = 5$ V, $V_H = 15$ V) for 15 min. A vortexlike flow behavior is observed as a consequence of the development of the EDL close to the membrane−anodic compartment interface. (b) Arrows representing the electrical field strength, superimposed on the velocity streamlines, in an area indicated by the dashed rectangle in (a). (c) Flow profile in the anodic compartment along the solid line in (a) after application of the external voltage ($V_L = 5$ V, $V_H = 15$ V) for 1, 7, and 15 min.

Table 2. Comparison with Other Work

ref	filter material	detection level	enrichment factor (time, min)	voltage (V_L, V_H) (V)
14	AA-*co*-HEMA hydrogel	fluorescein (5 μM)	100 (2)	100
16	porous titania membrane (sol−gel)	2,7-dichlorofluorescein (31 nM)	4×10^3 (7)	150, 300
20	surface-patterned Nafion	β-phycoerythrin (40 pM)	5×10^4 (5)	25, 50
22	nanochannel	rGFP (33 fM)	10^7 (40)	5, 10
23	nanofissures	FITC−DSA (100 pM)	10^3−10^5 (<10)	80, 80
this work	Nafion 117 strip	AF−BSA (60 pM)	2×10^4 (7)	5, 15

for various concentrations of AF−BSA within 10 min. Our device is capable of reaching a sensitivity as low as 60 pM for AF−BSA detection, which is among the best performing preconcentration devices. It has the potential to be conveniently integrated into a (portable) microfluidic analytical system thanks to the low-voltage operation. The use of a microfluidic preconcentrator device has already been shown to be at the basis of the enhancement of the analytical sensitivity of an enzyme-linked immunosorbent assay (ELISA): the limit of detection of prostate-specific antigen was decreased to a few picograms per milliliter, 2 orders of magnitude lower than the result obtained when no preconcentration device was used. In the future, this type of microfluidic preconcentration device may be essential in providing miniaturization and automatized handling and may offer new analytical possibilities for proteomic applications on single cells.

We also have numerically solved the electrokinetic model consisting of four coupled equations, i.e., the Poisson, Nernst−Planck (two equations for the two buffer ions), and Navier−Stokes (incorporated with the continuity equation) equations, which provided key insight into the mechanism of preconcentration. Our study has revealed that the permselective membrane-induced ion depletion at the anodic compartment side of the device gives rise to a secondary EDL and EOF of the second kind and simultaneously contributes to a local electrical field maximum. This balance between the locally strong electrophoretic force on the negatively charged tracer biomolecules and the combined EOF of the first and second kinds of the bulk fluid is responsible for

the biomolecule concentration in a part of the microfluidic channel, which is close to a vortexlike microfluidic pattern and where the electrical field is maximum, consistent with the experimental observations.

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SUPPORTING INFORMATION AVAILABLE

Additional information as noted in the text. This material is available free of charge via the Internet at http://pubs.acs.org.

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